

EDUCATION

Georgia Institute of Technology	Atlanta, United States
Ph.D. in Electrical and Computer Engineering (GPA: 4.00/4.00)	2024 - 2029 (Expected)
Seoul National University (SNU)	Seoul, South Korea
B.S. in Electrical and Computer Engineering (GPA: 4.24/4.30, Rank: 2/107)	2018 - 2024
<i>Summa Cum Laude</i>	

RESEARCH EXPERIENCE

FAST Lab, Georgia Tech (Advisor: Moinuddin Qureshi)	Jan 2025 - Current
LLM (Large Language Model) Offloading Scheme	
– Increases LLM inference throughput by offloading weights/KV cache to memory tiers outside GPU HBM.	
Reducing MoE (Mixture of Experts) LLM Inference Latency through Expert Reduction [1]	
– Developed a policy to decrease MoE decode latency by reducing the total number of activated experts.	
– Implemented inside vLLM, achieved up to 23% speedup with less than 1% accuracy loss.	
Increasing DRAM Bandwidth Utilization by Minimizing Bus Turnaround Overhead [2]	
– Analyzed the effect of different DRAM address mappings on delays between read/write commands.	
– Evaluated by modifying ChampSim. Achieved up to 11% speedup, and 3.5% on real systems.	
Scalable Computer Architecture Lab, SNU (Advisor: Jung Ho Ahn)	Jan-Feb, Jul-Dec 2023
Optimization of CNN Inference Latency within FHE (Fully Homomorphic Encryption) [3]	
– Proposed a state-of-the-art algorithm for evaluating CNN inference within FHE.	
– Implemented using C++, CUDA, and the HEAAN library. Achieved x1.5–2.7 speedup for ResNet18/50.	
GPU Acceleration of FHE	
– Implemented using C++ and CUDA. Achieved up to 10% speedup for a single ciphertext multiplication.	
Accelerated Intelligent Systems Lab, SNU (Advisor: Jinho Lee)	Mar-Jun 2023
DRAM PIM (Processing In Memory) Design and Evaluation [4][5]	
– Accelerates random access workloads by internally gathering scattered data.	
– Evaluated by modifying and integrating gem5 and Ramulator.	

PUBLICATIONS

- [1] V. Gupta, **J. H. Ju**, K. Sinha, A. Gavrilovska, and A. Iyer, “Lynx: Efficient moe inference through dynamic, batch-aware expert selection”, *in submission*,
- [2] H. Taneja, **J. H. Ju**, A. Saxena, and M. Qureshi, “Splitstream: Maximizing system bandwidth utilization via read-write buffer management”, *in submission*,
- [3] **J. H. Ju**^{*}, J. Park^{*}, J. Kim, M. Kang, D. Kim, J. H. Cheon, and J. H. Ahn, “Neujeans: Private neural network inference with joint optimization of convolution and FHE bootstrapping”, *ACM Conference on Computer and Communications Security*, 2024.
- [4] C. Shin, T. Kwon, J. Song, **J. H. Ju**, F. Liu, Y. Choi, and J. Lee, “A case for in-memory random scatter-gather for fast graph processing”, *IEEE Computer Architecture Letters*, 2024.
- [5] C. Shin, J. Song, H. Jang, D. Kim, J. Sung, T. Kwon, **J. H. Ju**, F. Liu, Y. Choi, and J. Lee, “Piccolo: Large-scale graph processing with fine-grained in-memory scatter-gather”, *IEEE International Symposium on High-Performance Computer Architecture*, 2025.
- [6] H. Kim, **J. H. Ju**, H. S. Choi, H. Roh, and W.-S. Choi, “Neuraleq: Neural-network-based equalizer for high-speed wireline communication”, *arXiv preprint arXiv:2308.02133*, 2023.

SCHOLARSHIPS AND AWARDS

Overseas PhD Scholarship, Korea Foundation for Advanced Studies (KFAS) Research grant \$13k/year	2024 - 2029
Presidential Science Scholarship, Korea Student Aid Foundation (KOSAF) Full tuition and stipend for eight semesters, total \$44k	2018 - 2023

WORK EXPERIENCE

Qualcomm Engineering Intern, On-target software team – Improving traceability of the Executorch AI compiler for edge devices – Designed and implemented the automatic tracing of torch.fx graph modifications with less than 3% overhead	May-Aug 2025
CryptoLab Research Intern, Core Development Team – Multi-GPU acceleration of Llama2 and ResNet18 within FHE. – Using C++ and CUDA, designed and implemented the initial structure of a new FHE library.	Apr-Jul 2024

TEACHING

Teaching Assistant for ECE 4100/6100 - Advanced Computer Architecture	Fall 2025
Teaching Assistant for ECE 4100/6100 - Advanced Computer Architecture	Spring 2025

RELEVANT COURSEWORK

- **Computer Architecture:** Hardware Software Co-Design for Machine Learning Systems, Advanced Computer Architecture, Computer Organization and Design
- **Systems:** Operating Systems, Systems Programming, Introduction to Data Communication Networks
- **Machine Learning:** Generative Deep Learning, Machine Learning Fundamentals
- **Digital Design:** Digital Systems Design and Experiments, Digital Integrated Circuits,

ACADEMIC PROJECTS

Linux Kernel: Custom Scheduler and Read/Write Lock – Implemented a WRR scheduler with periodic load balancing, and a range-based lock in the Linux kernel.	Spring 2023
FPGA CNN Accelerator – Using Verilog, implemented computation modules for a CNN accelerator on FPGA (Arty A7), and tested the inference its inference performance for a VGGNet variant.	Fall 2022
16 bit Pipelined CPU with Cache and DMA – Using Verilog, implemented a 16 bit pipelined CPU that supports a simplified MIPS ISA, with a write-back, write-allocate cache. Also implemented a simple DMA logic with cycle stealing.	Spring 2022

SKILLS

- **Programming:** C++, CUDA, C, Python, SystemVerilog, Matlab
- **Machine Learning Frameworks/Runtimes:** vLLM, Executorch, Pytorch
- **Architecture Simulators:** gem5, ChampSim, Ramulator, Scale-Sim, ASTRA-Sim
- **Tools:** Xilinx Vivado, Xschem, Ngspice, Magic VLSI

EXTRACURRICULAR ACTIVITIES

Nongnet Agricultural Commodity Price Prediction AI Competition Achieved a top 13% ranking out of 69 participating teams. Developed an AI model for price prediction utilizing a 10-year agricultural transaction database.	Fall 2022
7th Airforce Communication Service Group, Republic of Korea Air Force Crewman of the mobile TACAN (TACTical Air Navigation) system.	Nov 2019 - Jun 2021